B.M.S. COLLEGE OF ENGINEERING

Department of Computer Science and Engineering

Mobile App Development(23CS4AEMAD)

Synopsis Report

On

Attendance Tracker

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INRODUCTION

Attendance management is one of the essential academic activities in any institute to keep track of students' participation and academic regularity. However, traditional methods of attendance are manual, time-consuming, and inefficient. With increased usage of Web technologies, simple, interactive digital attendance system is being felt. The **Attendance Tracker** project is a mobile application developed using react native with a user-friendly interface for managing and viewing attendance. The system focuses on ease of use, cleanliness of design, and role-based interaction for students and teachers.

OBJECTIVE

The objective of this project is to develop a mobile-based Attendance Tracker application using React Native that enables efficient management of student attendance. The application aims to provide role-based access for Students and Teachers, where students can view their profile and attendance records, while teachers can mark attendance and review or update previously recorded attendance history. The system will store attendance data using a backend built with Node.js and MongoDB, ensuring reliable data management. The project focuses on creating a simple, user-friendly mobile interface with modular components that allow easy maintenance and future enhancements while improving the overall process of recording and tracking attendance.

FEATURES:

Key Features of the Attendance Tracker Mobile Application

- **Role-Based Authentication:** A single login screen allows both Students and Teachers to access the system based on their role.
- **Student Profile Management:** Students can view their personal details including Name, Gender, USN (Roll Number), and Class.
- **Subject-Wise Attendance Tracking:** Students can check attendance records for each subject individually.
- **Attendance Statistics:** The system displays present/absent records along with the overall attendance percentage for every subject.
- **Teacher Attendance Management:** Teachers can mark attendance for students belonging to the classes and subjects assigned to them.
- **Attendance Record Review:** Teachers can view previously recorded attendance details for verification.
- **Attendance Editing Capability:** Teachers can modify or update previously recorded attendance when required.

REQUIREMENTS:

Hardware Requirements

• A personal computer or laptop with the following minimum specifications or higher:
– Intel Core i3 processor or higher
– 8 GB RAM (minimum 4 GB supported for basic execution)
– 500 GB Hard Disk / SSD storage
– Standard keyboard and mouse

• An Android smartphone capable of running modern Android applications, used for testing and debugging the mobile application through USB debugging.

• A stable internet connection (2 Mbps or higher) for accessing development tools, libraries, and documentation.

Software Requirements

• **Operating System**
– Windows 10 / 11, Linux (Ubuntu or similar distributions)

• **Mobile Application Technologies**
– React Native
– JavaScript (ES6)
– React Native Components
– Draw.io (for diagrams and system design)

• **Development Environment / IDE**
– Visual Studio Code

• **Mobile Development Tools**
– Android Studio (for Android Emulator)
– Android SDK

• **Backend Technologies**
– Node.js
– Express.js

• **Database Technologies**
– MongoDB Atlas (Cloud Database)
– MongoDB Compass (Database GUI)

• **Server Environment**
– Node.js Runtime

• **Package Manager**
– npm (Node Package Manager)

• **Mobile Testing Platform**
– Android Emulator (Android Studio)
– Physical Android Device (USB Debugging)

• **Version Control**
– Github